altruistic animal has a cake to give to relatives, there is no reason at all for it to give every relative a slice, the size of the slices being determined by the closeness of relatedness. Indeed, this would lead to absurdity since all members of the species, not to mention other species, are at least distant relatives who could therefore each claim a carefully measured crumb! On the contrary, if there is a close relative in the vicinity, there is no reason to give a distant relative any cake at all. Subject to other complications like laws of diminishing returns, the whole cake should be given to the closest relative available. What I of course meant to say was 'We simply expect that second cousins should be as likely to receive altruism as offspring or siblings' (p. 122), and this is what now stands.

p. 122 He deliberately excludes offspring: they don't count as kin!

I expressed the hope that E. O. Wilson would change his definition of kin selection in future writings, so as to include offspring as 'kin'. I am happy to report that, in his *On Human Nature*, the offending phrase, 'other than offspring', has indeed—I am not claiming any credit for this!—been omitted. He adds, 'Although kin are defined so as to include offspring, the term kin selection is ordinarily used only if at least some other relatives, such as brothers, sisters, or parents, are also affected.' This is unfortunately an accurate statement about ordinary usage by biologists, which simply reflects the fact that many biologists still lack a gut understanding of what kin selection is fundamentally all about. They *still* wrongly think of it as something extra and esoteric, over and above ordinary 'individual selection'. It isn't. Kin selection follows from the fundamental assumptions of neo-Darwinism as night follows day.

p. 124 But what a complicated calculation …

The fallacy that the theory of kin selection demands unrealistic feats of calculation by animals is revived without abatement by successive generations of students. Not just young students, either. *The Use and Abuse of Biology*, by the distinguished social anthropologist Marshall Sahlins, could be left in decent obscurity had it not been hailed as a 'withering attack' on 'sociobiology'. The following quotation, in the context of whether kin selection could work in humans, is almost too good to be true:

> In passing it needs to be remarked that the epistemological problems presented by a lack of linguistic support for calculating *r*, coefficients of relationship, amount to a serious defect in the theory of kin selection. Fractions are of very rare occurrence in the world's languages, appearing in Indo-European and in the archaic civilizations of the Near and Far East, but they are generally lacking among the so-called primitive peoples. Hunters and gatherers generally do not have counting systems beyond one, two, and three. I refrain from comment on the even greater problem of how animals are supposed to figure out how that *r* [ego, first cousins] = ⅛.

This is not the first time I have quoted this highly revealing passage, and I may as well quote my own rather uncharitable reply to it, from 'Twelve Misunderstandings of Kin Selection':

> A pity, for Sahlins, that he succumbed to the temptation to 'refrain from comment' on how animals are supposed to 'figure out' *r*. The very absurdity of the idea he tried to ridicule should have set mental alarm bells ringing. A snail shell is an exquisite logarithmic spiral, but where does the snail keep its log tables; how indeed does it read them, since the lens in its eye lacks 'linguistic support' for calculating *m*, the coefficient of refraction? How do green plants 'figure out' the formula of chlorophyll?